MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT – 11** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (6 × 1 = 6 Marks)

Q1. As the acute angle θ increases from 0° to 90°, the value of $\sin\theta$: [1]

(A) Increases from 0 to 1 (B) Decreases from 1 to 0
(C) Remains constant (D) Decreases then increases

Q2. If $\tan\theta = \frac{4}{3}$, then the value of $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}}$ is: [1]

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{1}{2}$

Q3. The value of $(\sec^2\theta - 1)(\csc^2\theta - 1)$ is equal to: [1]

(A) 0 (B) 1 (C) -1 (D) 2

Q4. If $\cos\theta = \frac{1}{2}$ and $\sin\phi = \frac{1}{2}$ for acute angles θ and ϕ, then $\theta + \phi$ is: [1]

(A) 30° (B) 60° (C) 90° (D) 120°

Q5. If $a\cos\theta + b\sin\theta = 4$ and $a\sin\theta - b\cos\theta = 3$, then $a^2 + b^2$ is equal to: [1]

(A) 7 (B) 12 (C) 25 (D) 5

Q6. **Assertion (A):** The value of $\sin 60^\circ\cos 30^\circ + \sin 30^\circ\cos 60^\circ$ is 1.
Reason (R): $\sin 60^\circ = \frac{\sqrt{3}}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}, \sin 30^\circ = \frac{1}{2}, \cos 60^\circ = \frac{1}{2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. If $3\tan\theta = 4$, evaluate $\frac{3\sin\theta + 2\cos\theta}{3\sin\theta - 2\cos\theta}$. [2]

Q8. Prove that: $(1 + \cot^2\theta)(1 - \cos\theta)(1 + \cos\theta) = 1$. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Prove that: $\frac{1}{\csc A - \cot A} - \frac{1}{\sin A} = \frac{1}{\sin A} - \frac{1}{\csc A + \cot A}$. [3]

Q10. If $\cos\theta + \sin\theta = \sqrt{2}\cos\theta$, show that $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$. [3]

SECTION D: Long Answer / HOTS (2 × 5 = 10 Marks)

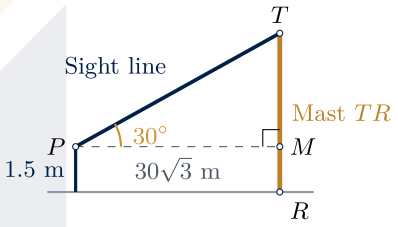
Q11. Prove that: $\frac{\tan^3\theta}{1+\tan^2\theta} + \frac{\cot^3\theta}{1+\cot^2\theta} = \sec\theta\csc\theta - 2\sin\theta\cos\theta$. [5]

Q12. If $\frac{\cos\alpha}{\cos\beta} = m$ and $\frac{\cos\alpha}{\sin\beta} = n$, show that $(m^2+n^2)\cos^2\beta = n^2$. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Theodolite Geodetic Surveying of a Telecom Mast

A surveyor sets up a digital theodolite at station P on level ground to measure the height of a cellular communication tower TR. The telescope of the theodolite is mounted on a tripod at height $PQ = 1.5$ m above the ground. The surveyor measures the angle of elevation to the top of the tower T to be $\theta = 30°$. The horizontal distance between the tripod and the tower is $d = 30\sqrt{3}$ m.



Based on the above geometric model, answer the following questions:

(i) Identify the right-angled triangle formed and write the relation for $\tan 30°$. [1]

(ii) Calculate the height of the tower section above the instrument level. [1]

(iii) Calculate the total height of the cellular tower from ground level, and find the direct line-of-sight distance from the theodolite telescope to the tower top T. [2]

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